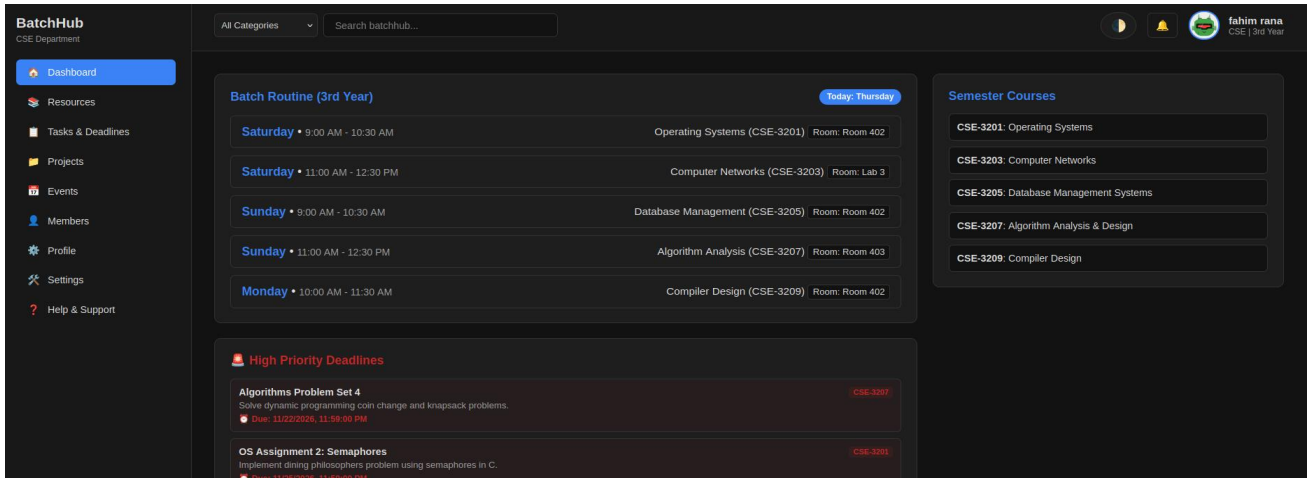
SOFTWARE DEVELOPMENT PROJECT

BatchHub

One Batch. One Space. Everything Connected.



Presented To:

Sharad Hasan
Lecturer,
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Presented by:

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Mustavi Mansur Tripto -- (2524203119)

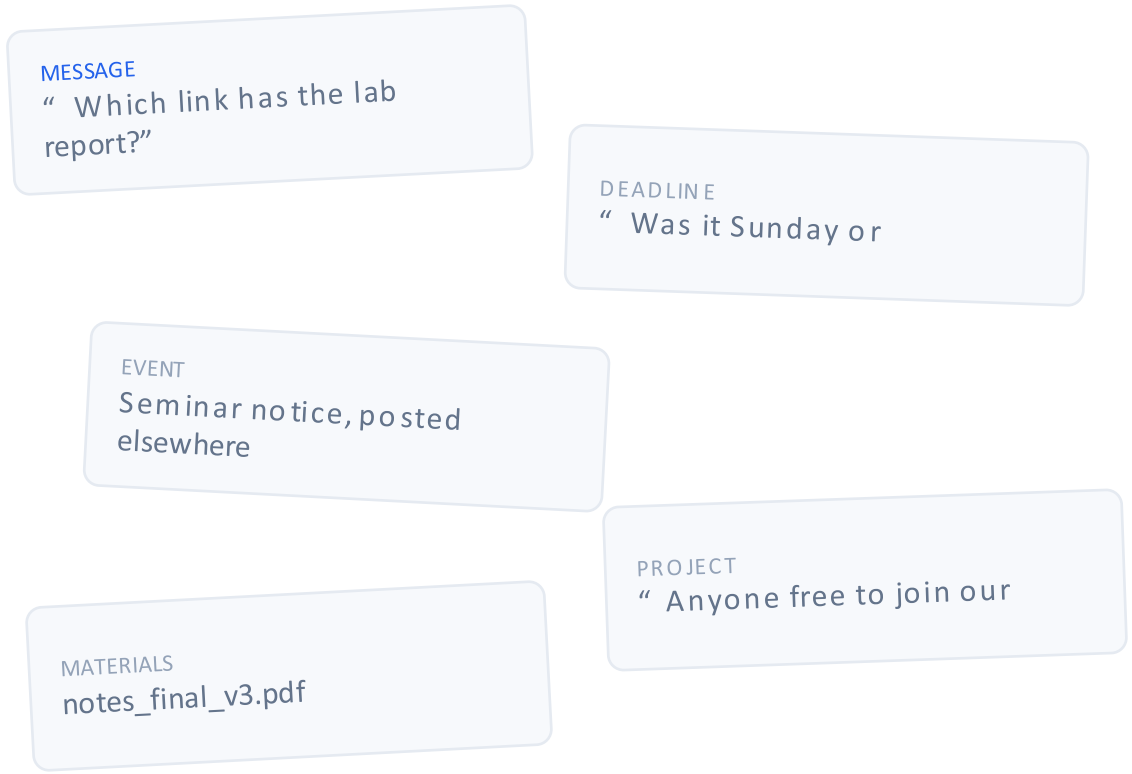
Fatema Tuj Johra Nepa -- (2524203163)

Tamima Ferdaous Amee -- (2524203123)

01 — **THE PROBLEM**

Academic Life Is Connected. Its Tools Aren't.

Important academic activities often become scattered across different communication channels and disconnected workflows.



01

Scattered communication

Important batch information can be distributed across different channels, making updates harder to track.

02

Disorganized academic tasks

Students need a clearer way to identify deadlines, monitor assignments and manage responsibilities.

03

Limited collaboration

Finding teammates, discovering projects and projects and coordinating participation can participation can become difficult.

04

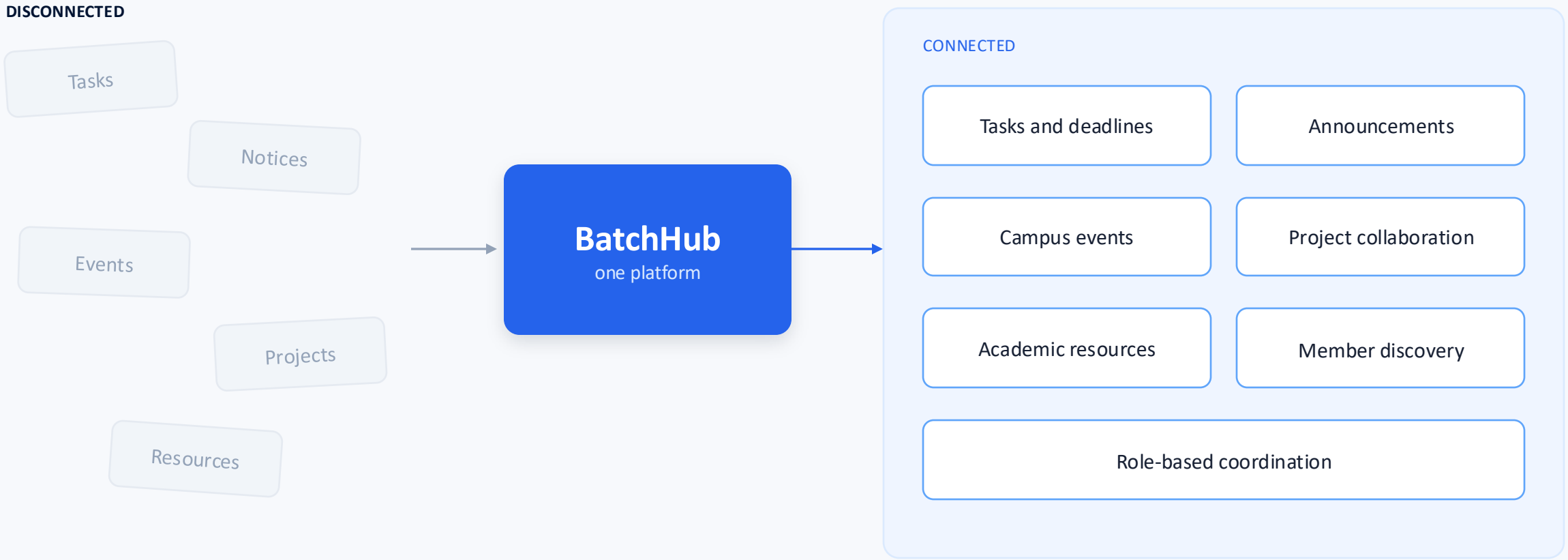
Unstructured resource sharing sharing

Notes, lab manuals, slides and learning materials need a more organized shared shared space.

02 — THE SOLUTION

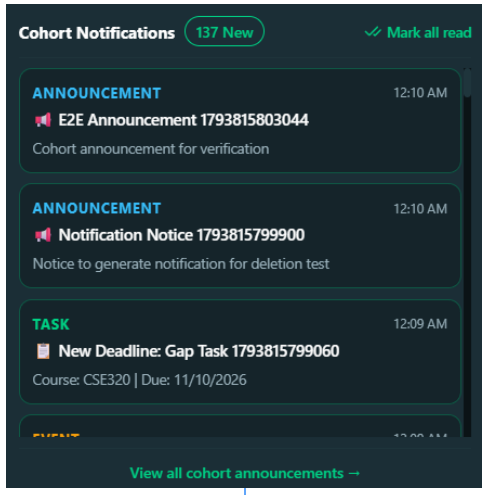
One Connected Space for the Entire Batch.

BatchHub brings academic coordination and student collaboration into one organized digital organized digital platform.

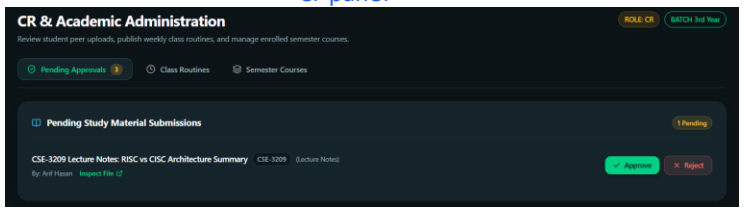


Everything Your Batch Needs. In One Place.

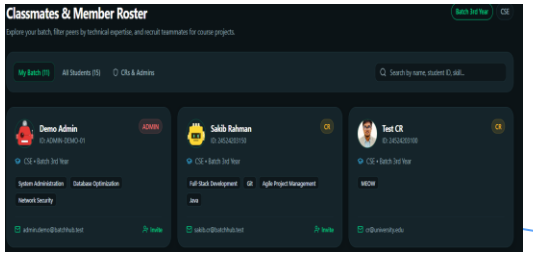
noticfications

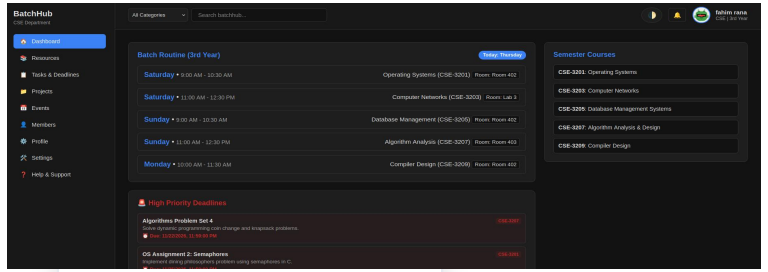


Cr panel

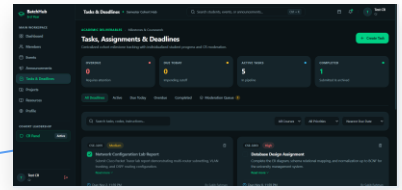


members

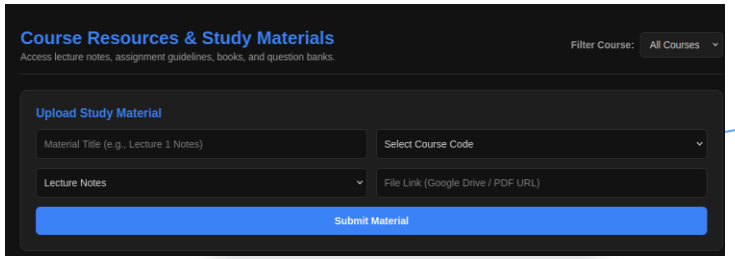




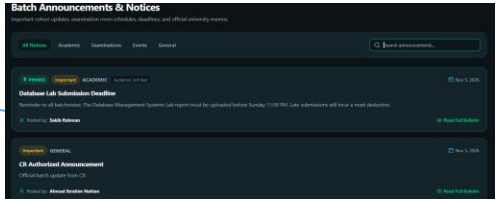
Dashboard



Tasks & Deadlines

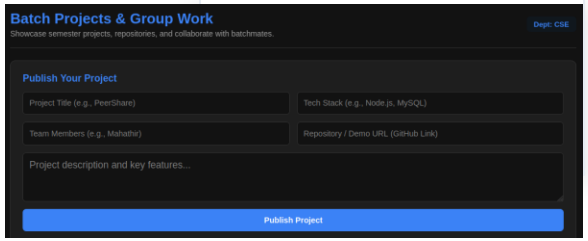


Resources

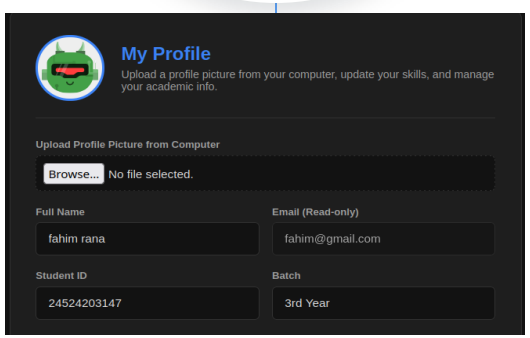


announcement

BatchHub



Projects



profile



Events

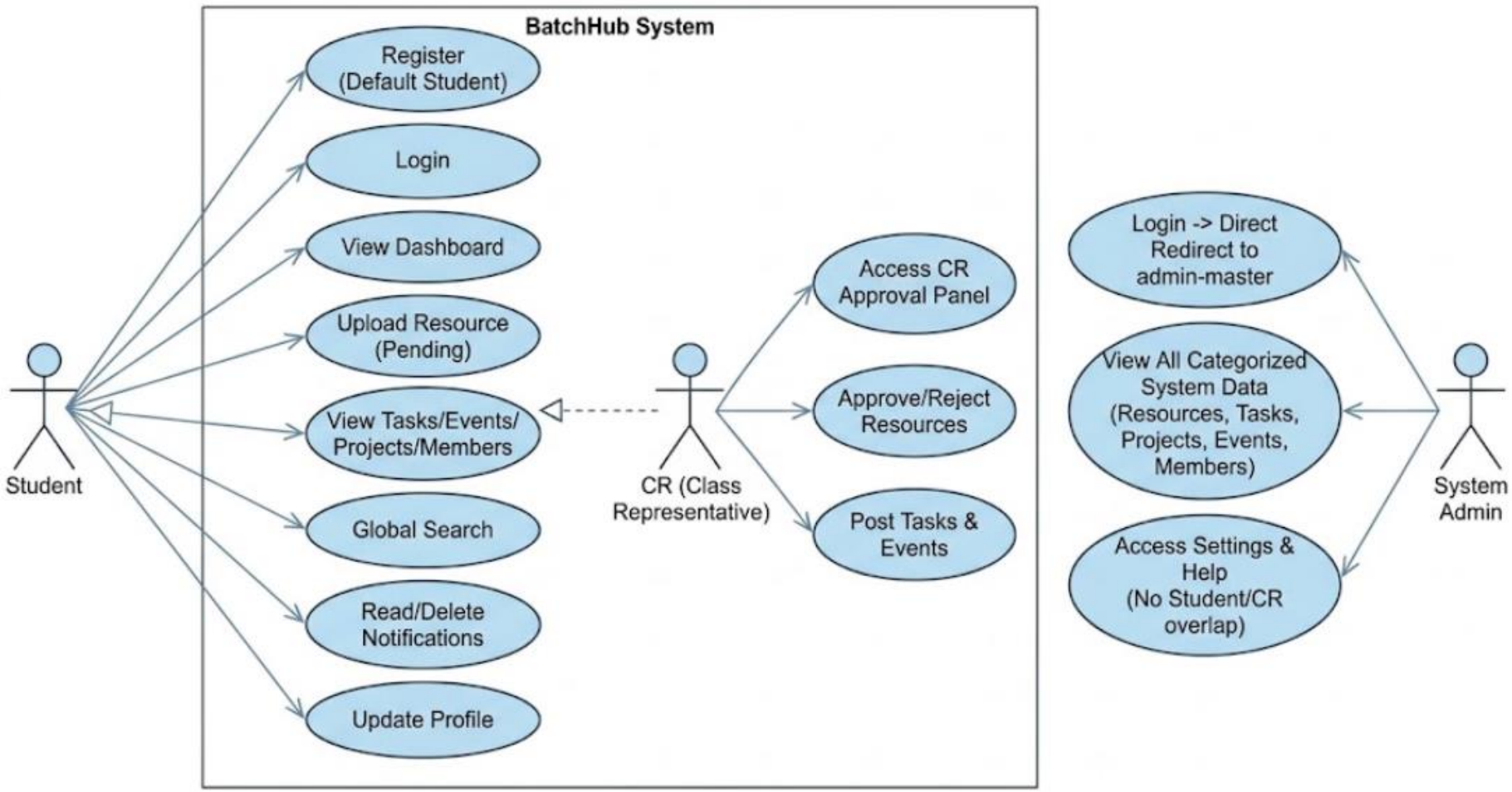


Figure 4.2: Use Case Diagram for Student, CR, and System Admin Roles

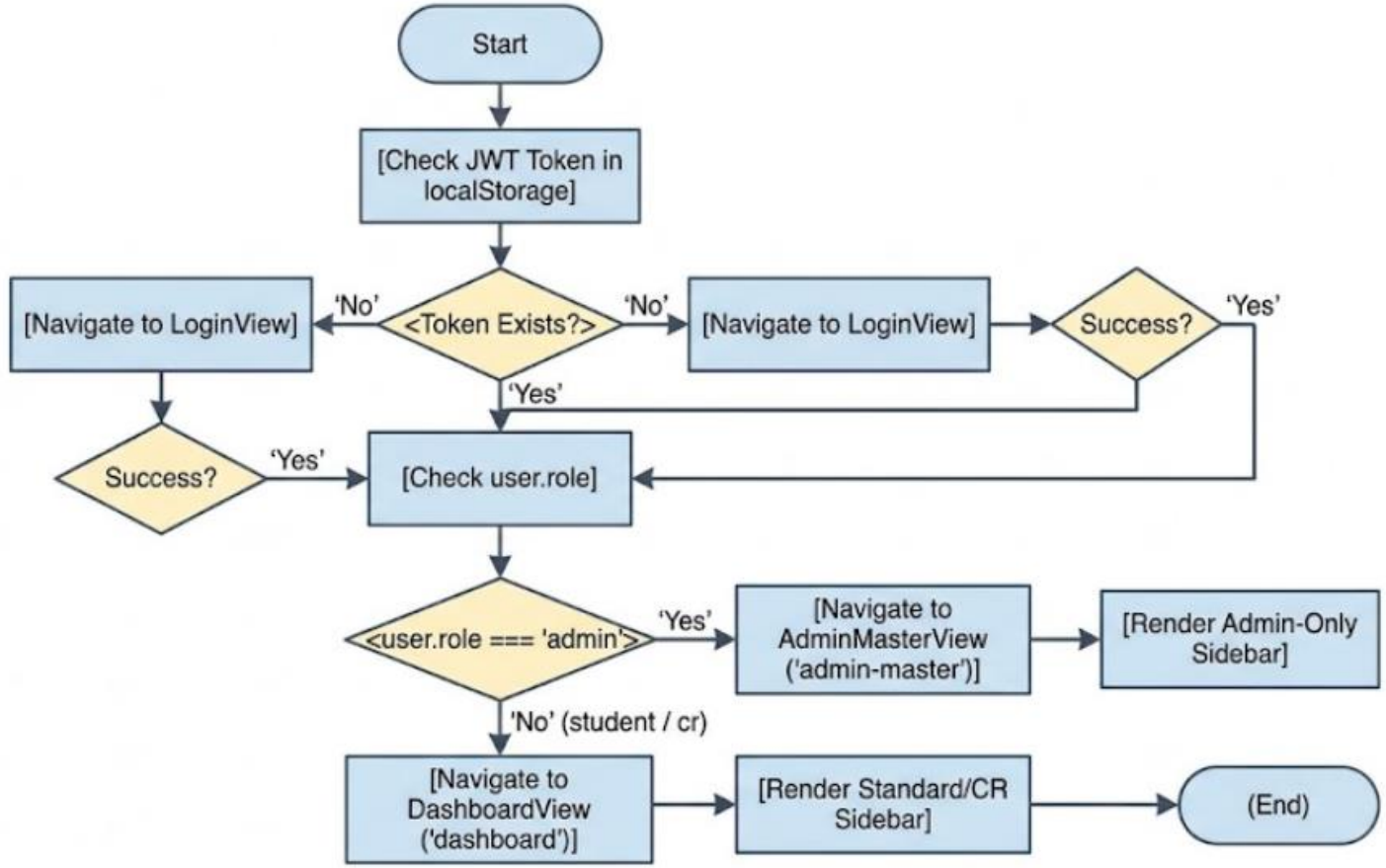


Figure 4.3: Role-Based Authentication and Routing Activity Flowchart

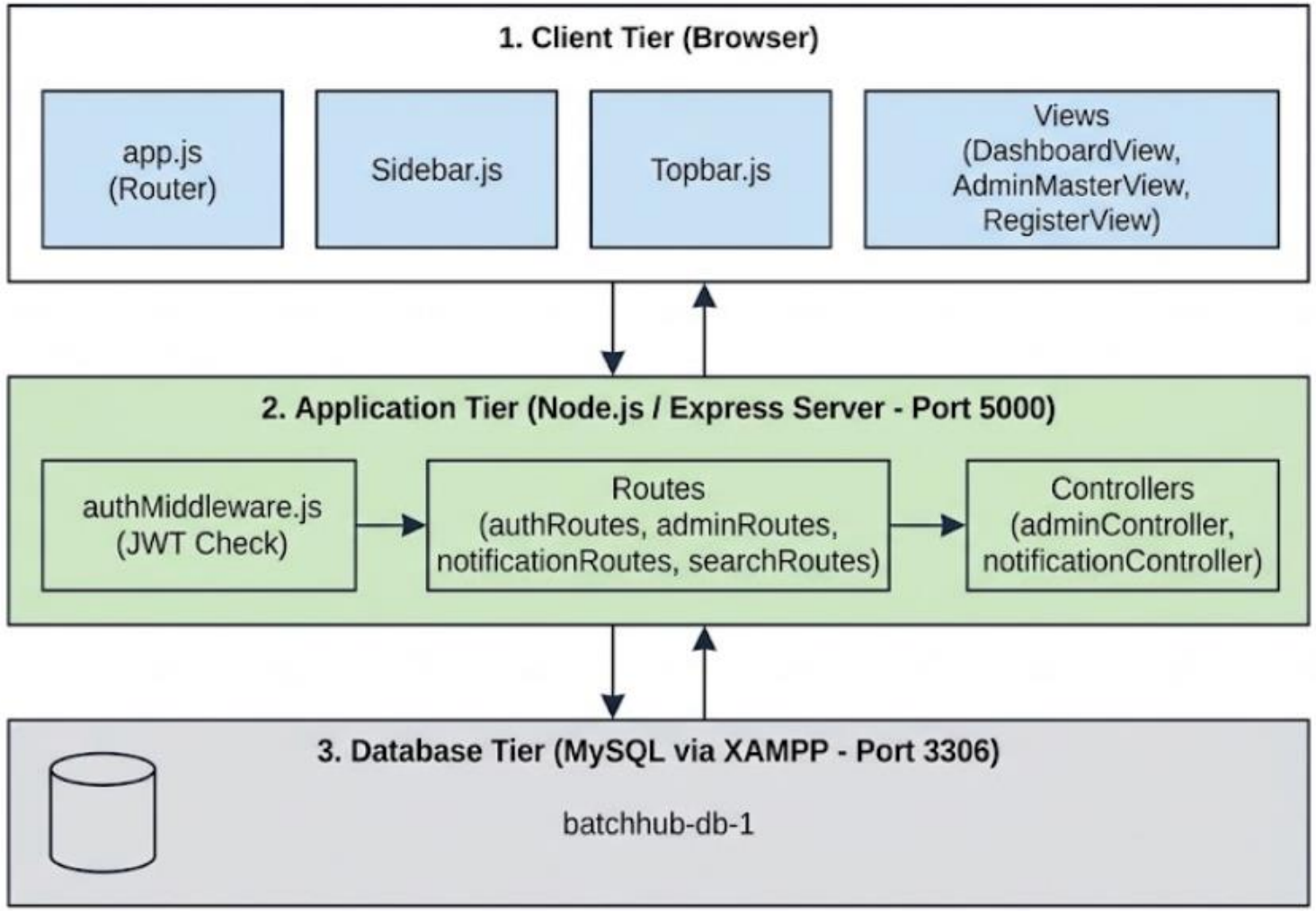
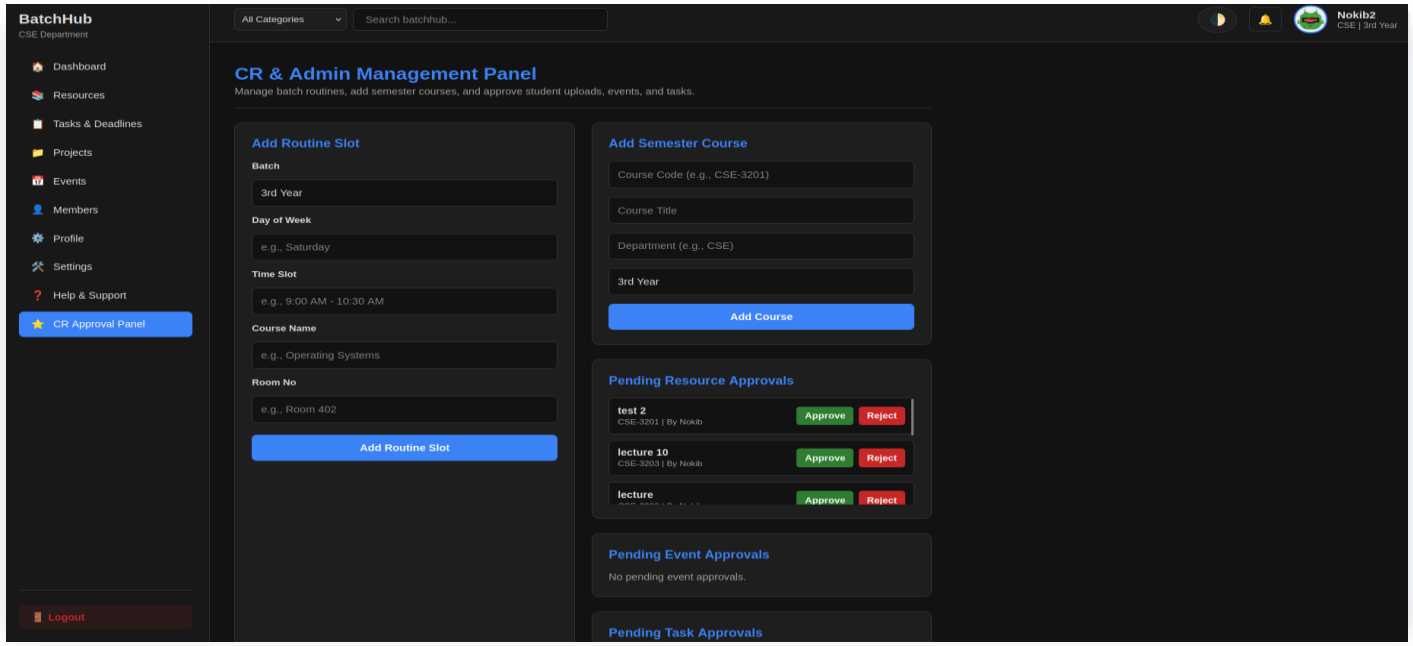


Figure 4.1: Three-Tier System Architecture of BatchHub

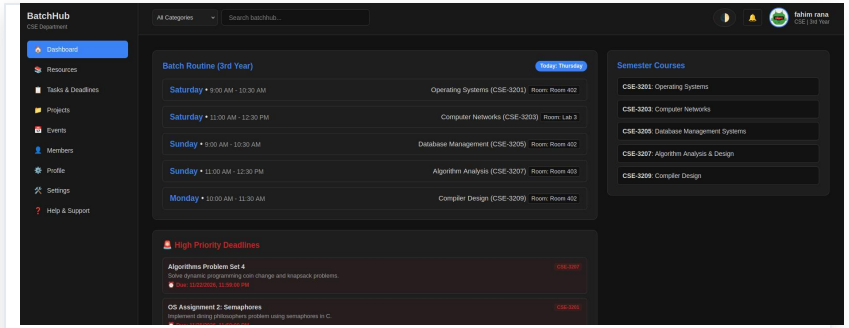
A Clear View of Your Academic Life.

The dashboard brings important academic information and batch activities into one accessible view.

CR DASHBOARD



STUDENT DASHBOARD



1 Academic overview

Overdue, completed and total task task metrics.

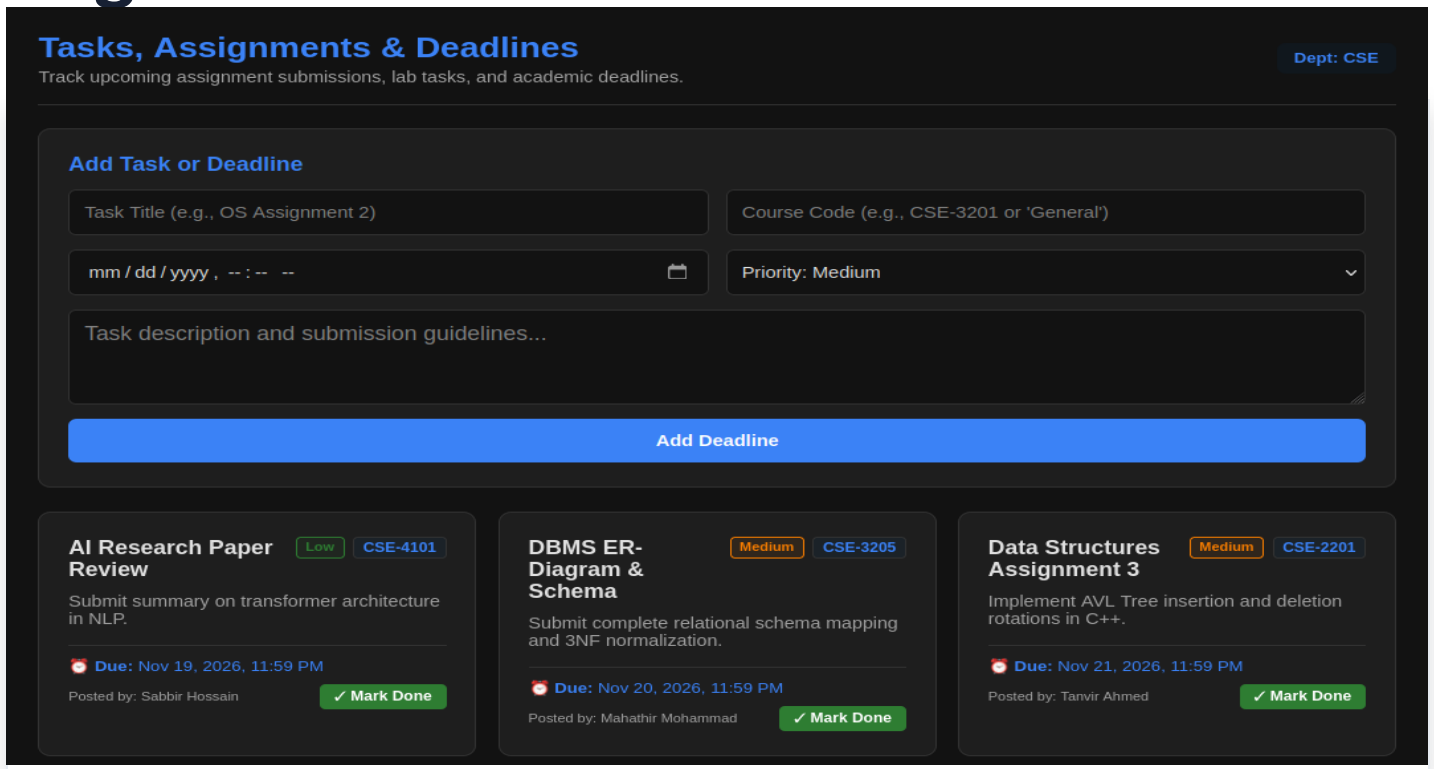
2 Tasks and notices

Deadlines and batch updates, one one click away.

05 — TASKS & ANNOUNCEMENTS

From Deadlines to Announcements. Everything Organized.

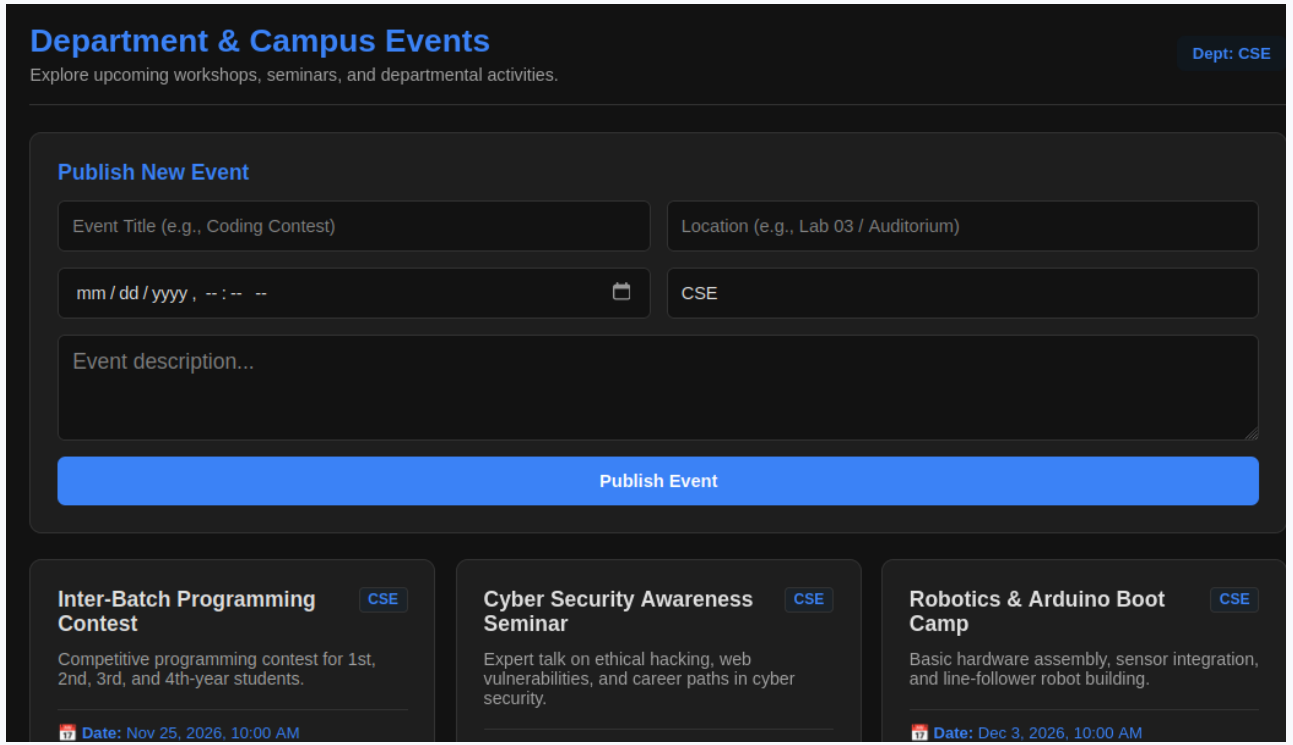
TASKS & DEADLINES



- Course code, priority and due date on every task
- Overdue, due today, active and completed states
- Filter by course, priority and due date

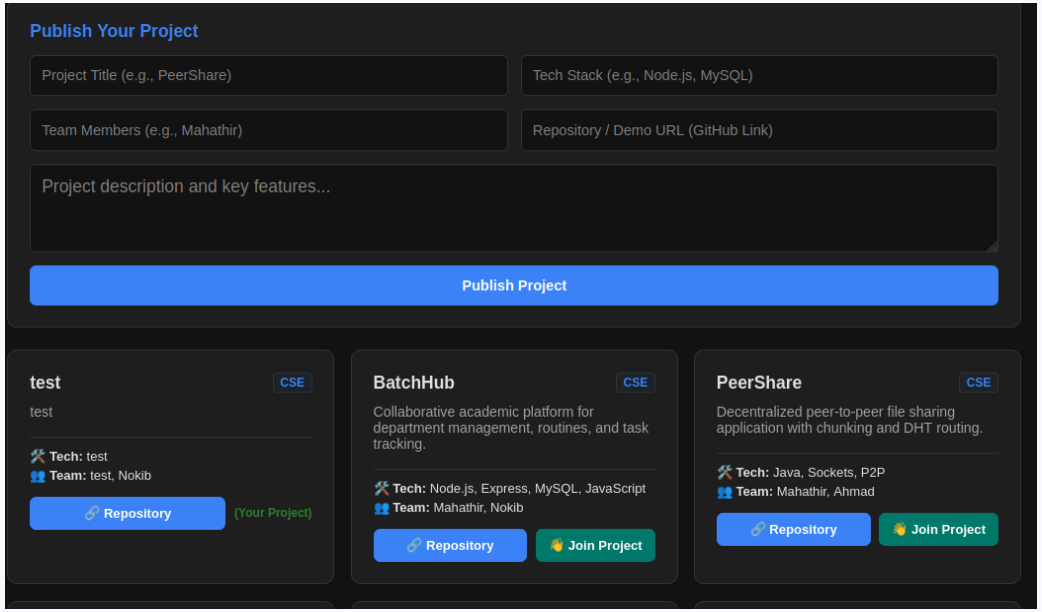
06 — EVENTS & PROJECTS

Where Academic Activities Meet Collaboration.



EVENTS

- Calendar-based discovery
- Upcoming and past events
- Venue and schedule details
- Event publishing



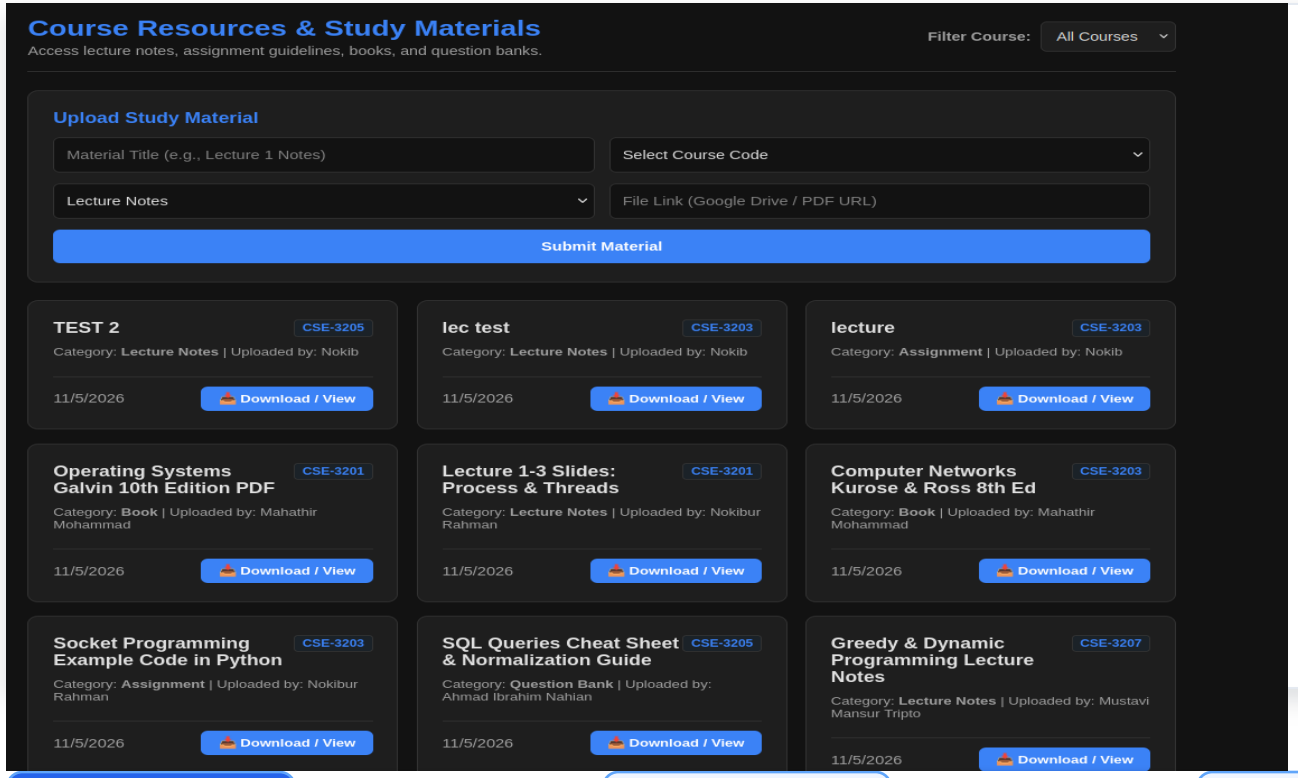
PROJECTS

- Project discovery and tech stacks
- Team info and join requests
- Project publishing

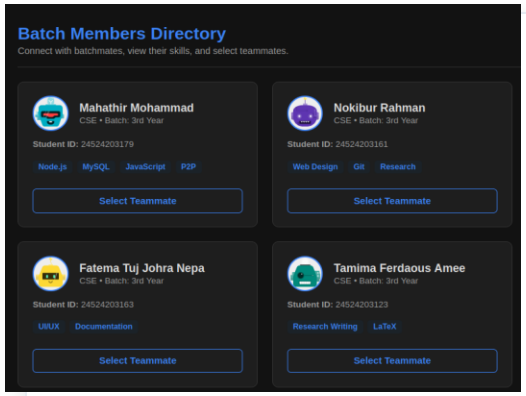
07 — RESOURCES & MEMBERS

Knowledge Should Be Easy to Find and Share.

RESOURCE LIBRARY



MEMBERS DIRECTORY



Upload form: title, course code, code, category and document link. link.

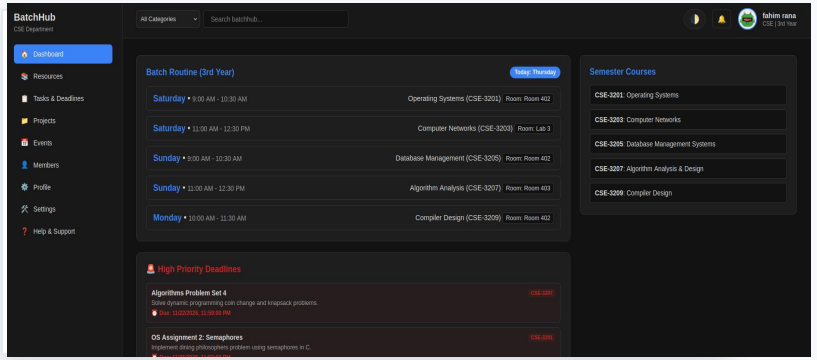
Discover Find classmates and and skills → **Share** Upload a document document link → **Organize** By course and category → **Access** Open shared materials

08 — ROLES

Different Responsibilities. One Shared Platform.

STUDENT

- Accesses academic information
- Manages tasks
- Discovers events
- Joins projects
- Accesses resources
- Maintains a profile



BatchHub

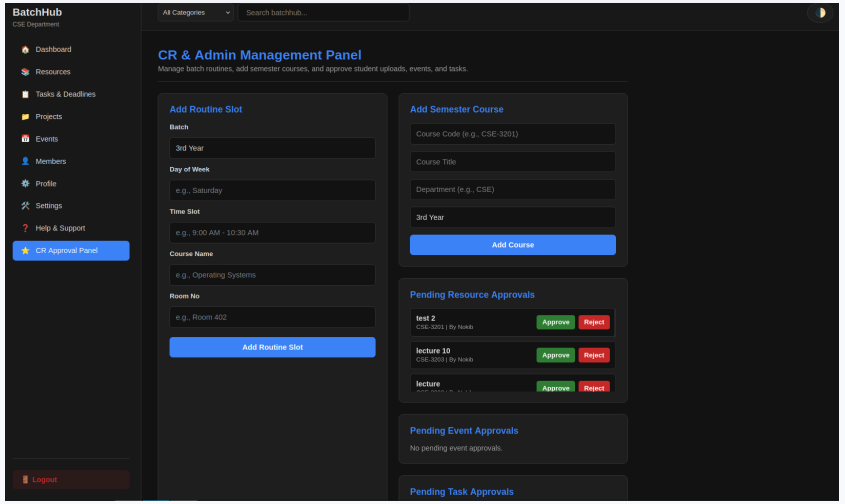
ADMIN

- Oversees platform administration
- Manages users and academic information
- Handles administrative responsibilities

CLASS REPRESENTATIVE (CR)

- Coordinates batch activities
- Reviews permitted submissions
- Manages relevant academic information
- Handles moderation workflows

Reviewed in the CR panel: study materials, event suggestions and task deadlines.



09 — CR ADMINISTRATION

Coordination With Structure and Accountability.

1

Student submission

Study material, event suggestion or task deadline

2

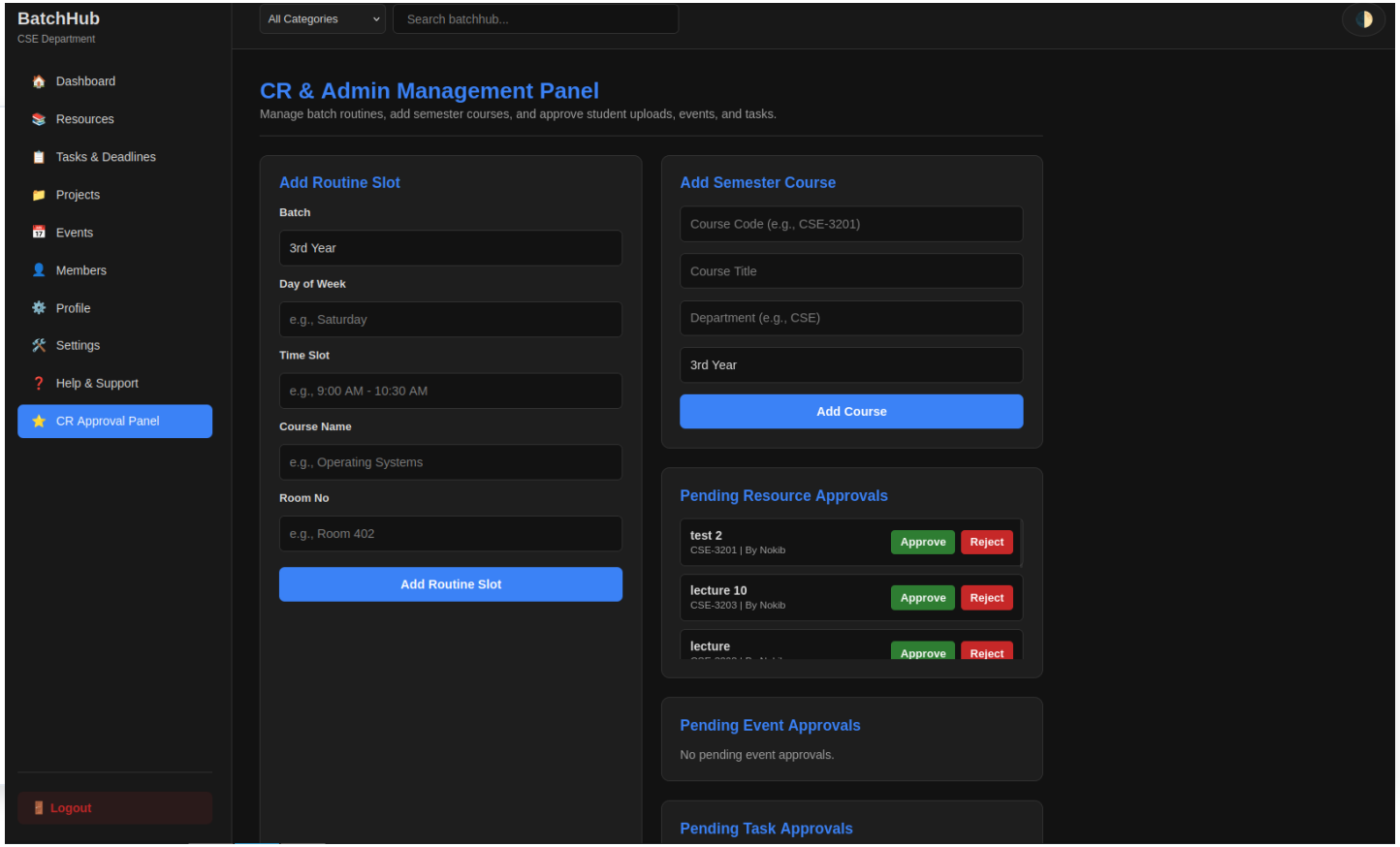
Review queue

Pending Approvals tab in the CR panel

3

Approve / Reject

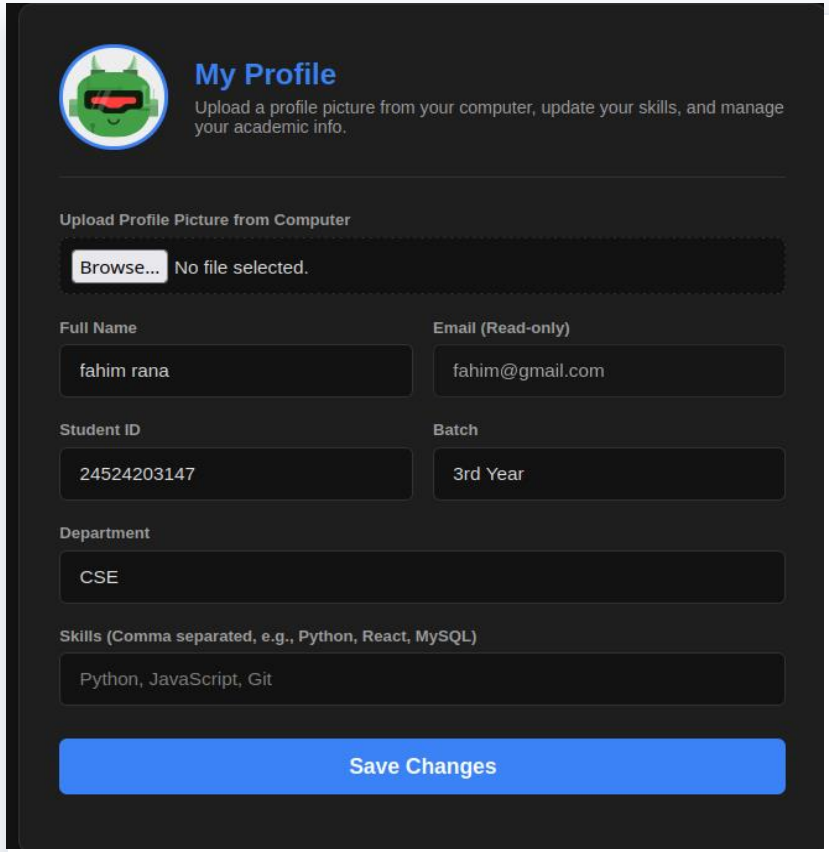
CR decides on each item



CREATION FORMS — EVENT · ANNOUNCEMENT · TASK · RESOURCE

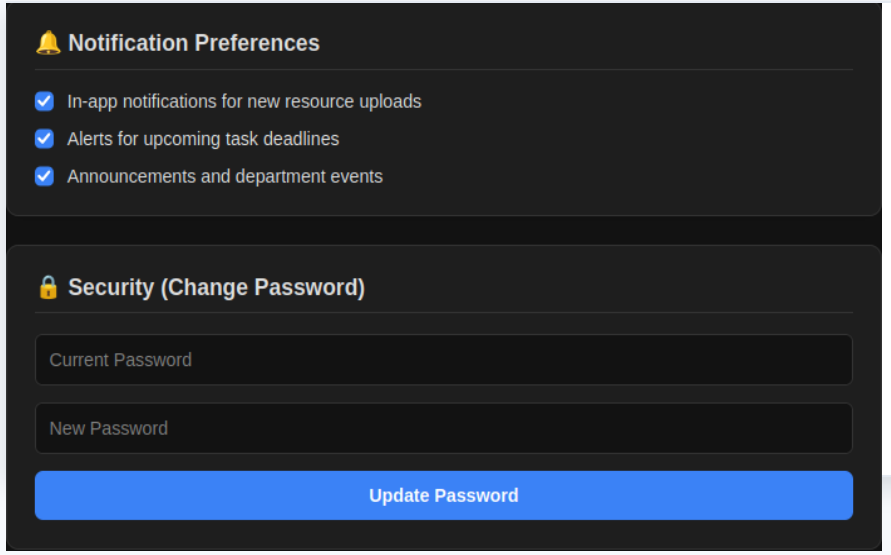
A More Organized Personal Experience.

PERSONAL PROFILE

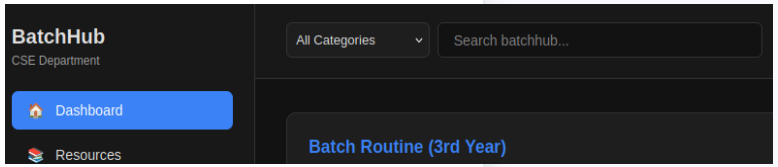


Academic identity, student information and technical skills shown across cohort directories.

ACCOUNT SETTINGS



QUICK NAVIGATION



Global search with shortcuts to Dashboard, Tasks, Tasks, Events and Announcements.

Under the Interface. A Connected System.

1 Frontend

React 19 · TypeScript · Vite · Tailwind CSS v4

Renders the interface, handles interactions, presents data.

2 REST API

Requests and responses (JSON)

The frontend talks to the backend over REST.

3 Backend

Node.js · Express.js

Application logic and API operations.

4 Database

MySQL

Stores the application's persistent persistent information.

SUPPORTING CONCEPTS

JWT
Authentication

Role-aware UI
CR Panel locked for Students

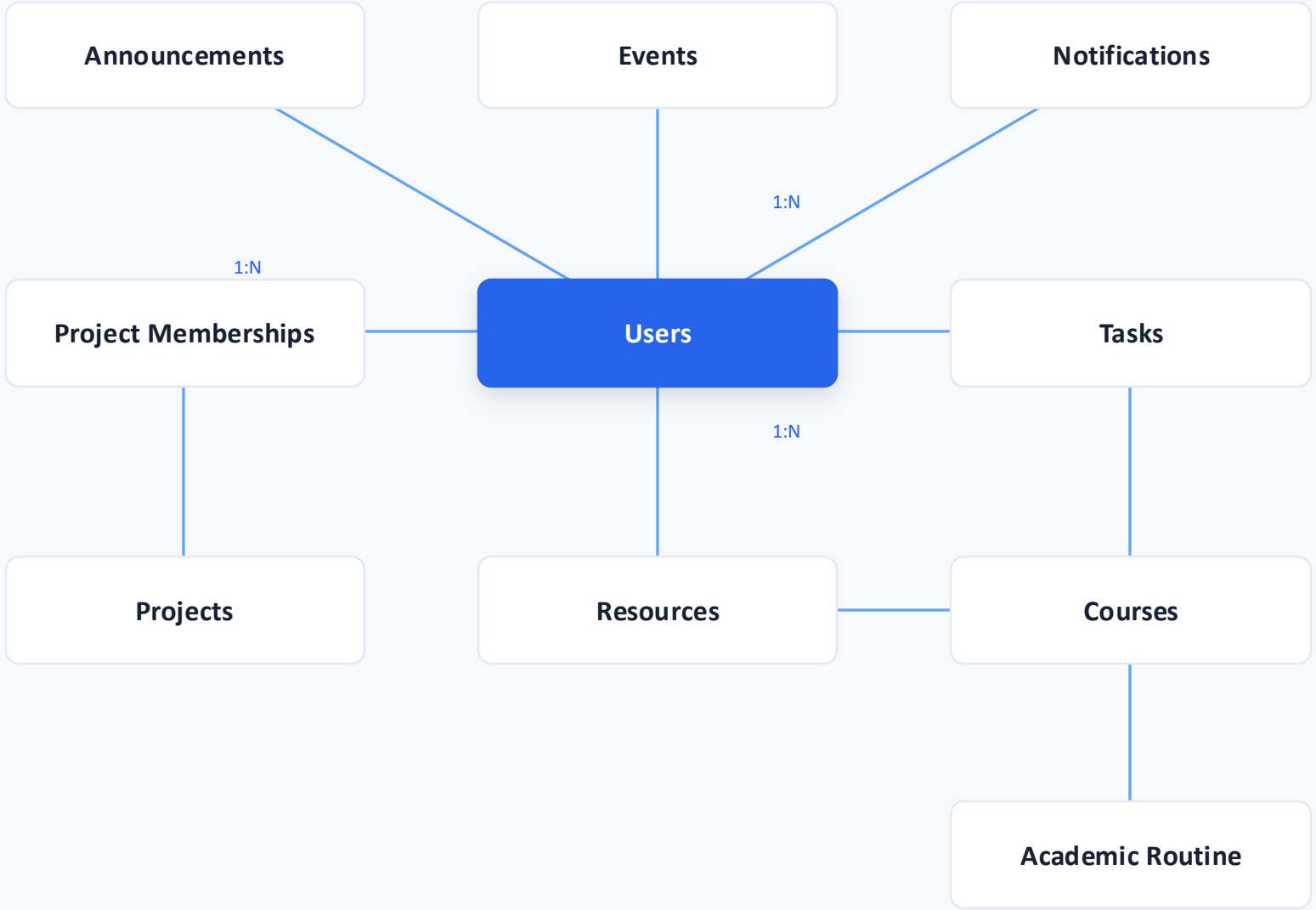
Zustand
Client-side state

TanStack Query
Server data fetching and caching

Structured Data. Connected Information.

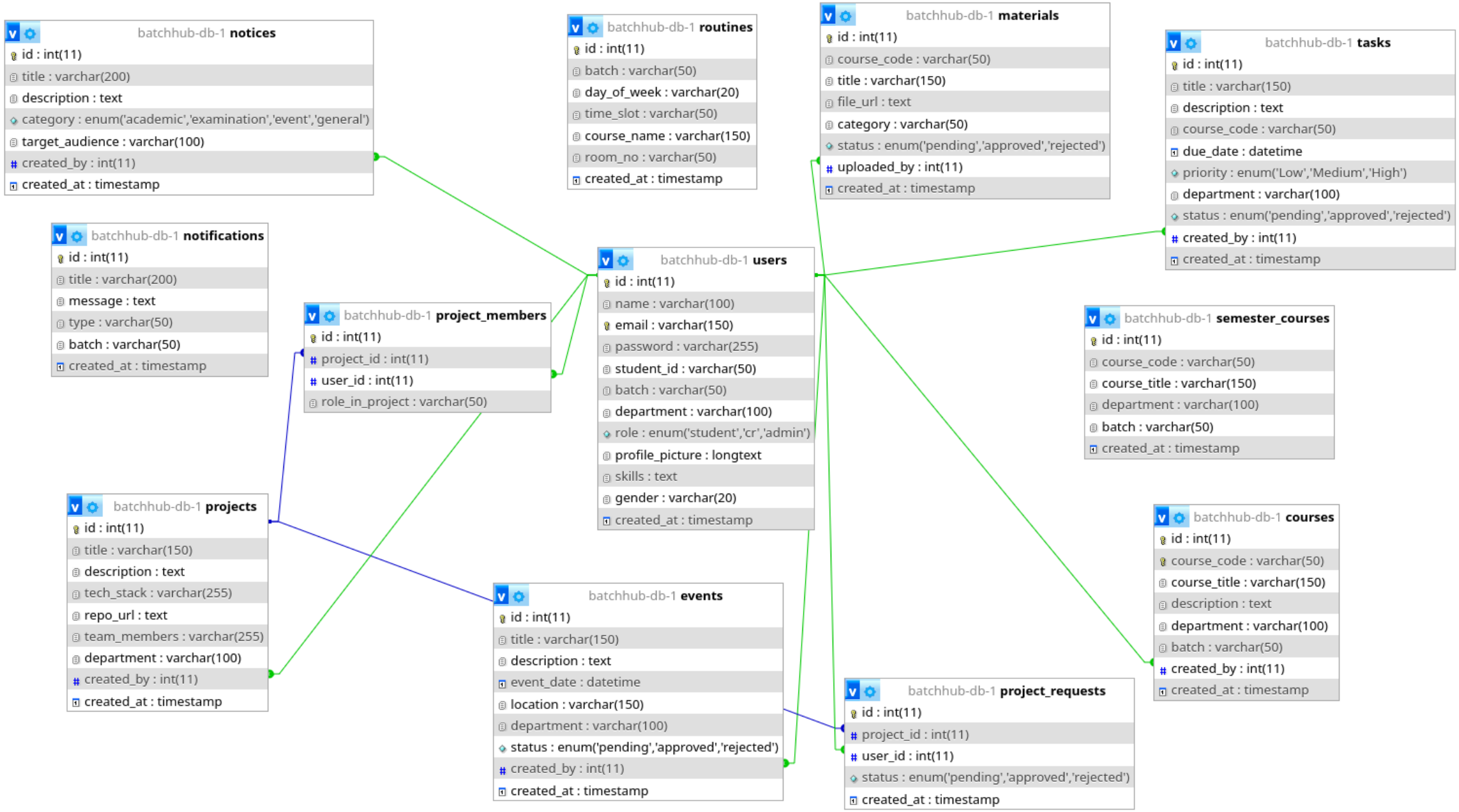
How the platform’s information is organized across modules.

- Users connect to what they create and join join
- Courses tie tasks, resources and routines together
- Submissions carry a review status



CONCEPTUAL DATA MODEL · NOT THE EXACT SCHEMA

ER Diagram (Entity Relationship Diagram)



Built Around Real Academic Workflows.

IMPLEMENTED

- **Frontend–backend separation**
  React UI talks to an Express API over REST
- **Component-based interface**
  Reusable React and TypeScript views
- **JWT authentication**
  Token-based sign-in
- **Role-aware access**
  Student, CR and Admin see different tools
- **Managed state and data**
  Zustand for client state, TanStack Query for server data
- **Form-based data entry**
  Required fields and a minimum password length in forms

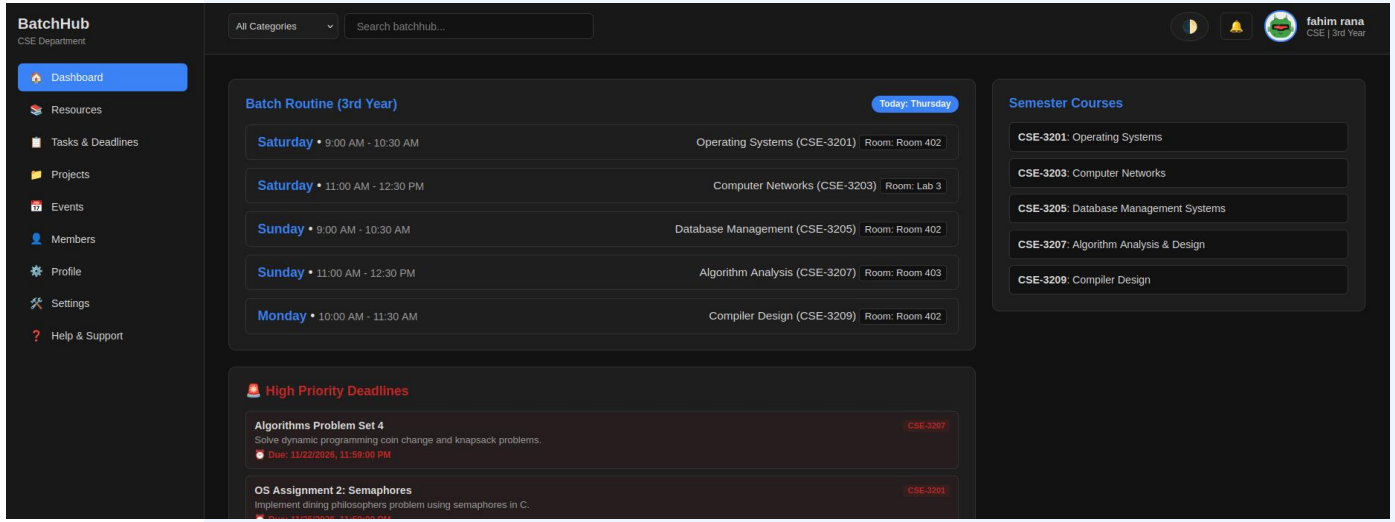
FUTURE SCOPE · NOT YET IMPLEMENTED

- Dedicated mobile application
- Real-time communication
- Advanced academic analytics
- Additional academic integrations
- Enhanced notification capabilities
- Further improvements to collaboration workflows workflows

ACADEMIC COLLABORATION, ORGANIZED.

Bringing the Batch Together.

BatchHub transforms disconnected academic activities into one organized collaborative experience.



BATCHHUB

One Batch. One Space. Everything Connected.

Thank You | Q&A